腦波 × 感應飛鏢 × AI

Brain signals × Sensor darts × AI

實驗原型 | Research prototype

01 核心概念 | The simple idea

量度訊號 → 玩遊戲 → 睇回饋



The simple idea

Measure signals → Play → Get feedback



02 兩路輸入 | Two sensor inputs

Muse 2：量度腦電訊號（EEG）
FSR：受壓時電阻改變
Arduino + 藍牙模組：讀取及傳送

位置靠感應器佈局；力度要校準



Two sensor inputs

Muse 2: measures EEG
FSR: resistance changes with force
Arduino + Bluetooth module: read and send

Location needs sensor layout; force needs calibration

03 即時遊戲 | Live game loop

Python：同步 → 去雜訊 → 分段
先靜坐量度，再投／輕按
遊戲：計分、提示、記錄

腦波 ≠ 直接讀心或專注度



Live game loop

Python: sync → clean → window
Measure while still, then throw / press
Game: score, prompt, log

EEG ≠ mind reading or direct attention

04 點樣整？ | How we build it

- EasyEDA：設計電路
- Autodesk Fusion：外殼同支架
- VS Code / Cursor：寫 Python、Arduino 程式

AI agent 草擬 → 人手審查 → 測試 → GitHub

How we build it

- EasyEDA: circuit design
- Autodesk Fusion: enclosure and mount
- VS Code / Cursor: Python and Arduino code

AI agent drafts → Human review → Tests → GitHub

05 額外 AI 層 | Optional AI layer

- Obsidian：整理筆記
- 文字分塊 + Embedding：方便搵尋
- RAG：先攞資料，再回答
- GraphRAG：加埋資料之間嘅關係
- Ollama：本地生成摘要；Slack：團隊通知

唔放入即時計分主線

Optional AI layer

- Obsidian: organise notes
- Text chunks + embeddings: enable search
- RAG: retrieve sources, then answer
- GraphRAG: also use linked relationships
- Ollama: local summaries; Slack: team alerts

Keep outside real-time scoring

06 點知 AI 準唔準？ | How do we test AI?

- 真實標籤 vs 預測 → 混淆矩陣
- 過擬合：舊數據好，新數據差
- 欠擬合：新舊數據都差
- 用未見過嘅人／時段測試

MoE：路由器揀專家；初版唔需要

How do we test AI?

- True labels vs predictions → Confusion matrix
- Overfitting: good on training, poor on new data
- Underfitting: poor on both
- Test on unseen people / sessions
- Nest on unseen people / sessions

MoE: router selects experts; not needed for v1

Do not imply separate sensor pipelines alone constitute MoE.

07 可以研究，唔好誇大 | Explore, do not overclaim

- 專注遊戲：研究參與度，唔係治療承諾
- 活動受限：可改輕按玩法，先由治療師評估
- Muse 2 唔能夠保證讀到運動意圖
- 唔取代 ADHD 治療或專業復康

最小可行版本：FSR + Arduino → Python → 計分，再加 EEG

Explore, do not overclaim

- Attention game: study engagement, not a treatment promise
- Limited movement: consider gentle presses after therapist assessment
- Muse 2 cannot guarantee motor-intent decoding
- Not a replacement for ADHD care or rehabilitation

MVP: FSR + Arduino → Python → Scoring, then add EEG

Evidence: JAMA Psychiatry neurofeedback meta-analysis (2025); Interlink FSR documentation